

Quiz: Investment Appraisal - Version B

FAST Lahore – Data Science & Cloud Computing Center

Name: _____

Roll No: _____

Scenario:

FAST National University, Lahore Campus, is planning to establish a new **Data Science & Cloud Computing Center**. The administration has compiled the following cost and revenue projections:

Initial Investment (Year 0):

Cost Component	Amount (PKR)
Land Acquisition & Site Preparation	50,000,000
Construction & Civil Works	135,000,000
Cloud Server Infrastructure & Data Center Equipment	75,000,000
Smart Classroom Technology & Furniture	30,000,000
Networking Infrastructure & Cooling Systems	20,000,000
Total Initial Investment	310,000,000

Projected Annual Cash Inflows (PKR):

The center will generate revenue through increased student enrollment, corporate training programs, cloud services to local businesses, and research collaborations.

Year	Tuition & Fees	Executive Programs	Research Grants	Industry Partners	Total Inflow
1	40,000,000	15,000,000	10,000,000	7,000,000	72,000,000
2	52,000,000	22,000,000	18,000,000	14,000,000	106,000,000
3	60,000,000	28,000,000	25,000,000	22,000,000	135,000,000
4	65,000,000	32,000,000	30,000,000	28,000,000	155,000,000
5	68,000,000	35,000,000	35,000,000	32,000,000	170,000,000
6	70,000,000	38,000,000	38,000,000	34,000,000	180,000,000

Annual Operating Costs:

Note: Operating costs are incurred every year from Year 1 through Year 6 (including the final year when equipment is sold).

Cost Item	Annual Amount (PKR)
Faculty & Staff Salaries	25,000,000
Utilities & Cooling	8,000,000
Software Licenses & Cloud Services	10,000,000
Administrative Overhead	5,000,000
Total Annual Operating Costs	48,000,000

Additional Information:

- Discount rate: **12%**
- At the end of Year 6, the server equipment will have a resale value of **PKR 22,000,000**
- Assume all cash flows occur at year-end

Required:

Part A (10 marks): Calculate the **Net Cash Flows** for each year (Year 0 through Year 6), including the resale value in the final year. Present your answer in a table.

Solution A: Net Cash Flows () indicate negative value

Year	Initial Investment	Inflows	Operating Costs	Resale Value	Net Cash Flow
0	(310,000,000)	-	-	-	(310,000,000)
1	-	72,000,000	(48,000,000)	-	24,000,000
2	-	106,000,000	(48,000,000)	-	58,000,000
3	-	135,000,000	(48,000,000)	-	87,000,000
4	-	155,000,000	(48,000,000)	-	107,000,000
5	-	170,000,000	(48,000,000)	-	122,000,000
6	-	180,000,000	(48,000,000)	22,000,000	154,000,000

Part B (20 marks): Using the Discounted Cash Flow (DCF) method, compute the **Present Value** of each year's net cash flow using the 12% discount rate. Present your calculations in a table showing: Year, Net Cash Flow, Discount Factor (formula: $1 \div (1+r)^t$), and Present Value.

Solution B: Present Value Calculation

Year	Net Cash Flow	Discount Formula (12%)	Present Value
0	(310,000,000)	-	(310,000,000.00)
1	24,000,000	$1 / (1 + 0.12)^1 = 0.892857143$	21,428,571.43
2	58,000,000	$1 / (1 + 0.12)^2 = 0.797193878$	46,237,244.92
3	87,000,000	$1 / (1 + 0.12)^3 = 0.711780248$	61,924,881.57
4	107,000,000	$1 / (1 + 0.12)^4 = 0.635518078$	68,000,434.34
5	122,000,000	$1 / (1 + 0.12)^5 = 0.567426856$	69,226,076.43
6	154,000,000	$1 / (1 + 0.12)^6 = 0.506631121$	78,021,192.63

Part C (5 marks): Calculate the **Payback Period** using cumulative net cash flows. Express your answer in years and months.

Solution C: Payback Period

Cumulative Cash Flows:

Year 0: (310,000,000)
Year 1: (286,000,000)
Year 2: (228,000,000)
Year 3: (141,000,000)
Year 4: (34,000,000)
Year 5: 88,000,000

Payback occurs during Year 5.

Fraction of Year 5 = 34,000,000 / 122,000,000 = 0.2787 years

Months = 0.2787 × 12 = 3.34 months

Payback Period = 4 years and 3.34 months (approx. 4 years and 3 months and 10 days or 51.34 months)

Part D (10 marks): Calculate the **Net Present Value (NPV)** of the project. Based on your calculation, should FAST Lahore proceed with the construction? Justify your recommendation using the NPV decision rule.

Solution D: Net Present Value (NPV)

Sum of Present Values (Years 1 to 6) = 21,428,571.43 + 46,237,244.92 + 61,924,881.57 + 68,000,434.34 + 69,226,076.43 + 78,021,192.63 = 344,838,400.89

NPV = Total PV of Cash Inflows - Initial Investment

NPV = 344,838,401.32 - 310,000,000.00 = **PKR 34,838,401.32** (or if no decimal values are considered 34,838,398). Any value within the range is okay as each person may have rounded at different decimal value.

Recommendation: Since the NPV is positive (PKR 34,838,401.32 > 0), FAST Lahore should proceed with the establishment of the Data Science & Cloud Computing Center.

Total Marks: 45